

AI-ASSISTED MAILBOX TRIAGE PILOT SIMULATION

Six-Stage Governance & Implementation Workflow

Pilot Definition through Final Evaluation — with Six-Month Shadow Operation

Read-only · Non-operative · Shadow-mode · No mailbox actions permitted

Process Overview

This document presents a desktop simulation of a small-scale, read-only AI-assisted mailbox triage pilot in an academic institution.

The initial use case was deliberately simple: allow an AI system to review incoming messages, suggest classifications, and compare those suggestions with human judgment without moving, modifying, or sending email. During the simulation, however, this limited use case developed into a much more complex workflow once mailbox permissions, privacy, records management, data minimization, model controls, human review, monitoring, stop conditions, and final decision-making were considered.

The final model maps that complexity into a six-stage governance and implementation workflow. It moves from pilot definition through method design, offline feasibility testing, environment build-out, pre-live verification, and a six-month shadow operation, concluding in a final evaluation with five possible outcomes.

Stage 4 comprises four coordinated workstreams that proceed together—identity and permissions, Azure and Microsoft 365 setup, data minimization and model controls, and reviewer interface and monitoring—rather than as a fixed sequence. Stage 6 is presented through two linked diagrams: an illustrative six-month timeline with concurrent review, and a separate stop-response process that can interrupt the timeline at any point.

The flowchart boxes are intentionally brief. Full activity details, ownership, outputs, and decision logic are documented in the reference tables that follow each diagram.

Diagram Parts

Part 1 — Stages 1–3: Define the Pilot, Design Method, Offline Feasibility Test (Gates 1–2)

Part 2 — Stage 4 coordinated workstreams (4A–4D) and Stage 5 readiness decision (Gate 3)

Part 3 — Stage 6 timeline: illustrative six-month sequence and concurrent review

Part 4 — Stage 6 stop-response diagram: must-not list, stop conditions, response chain

Part 5 — Final Evaluation and Gate 4: final decision (five branches)

Workflow Diagram — Part 1 of 5

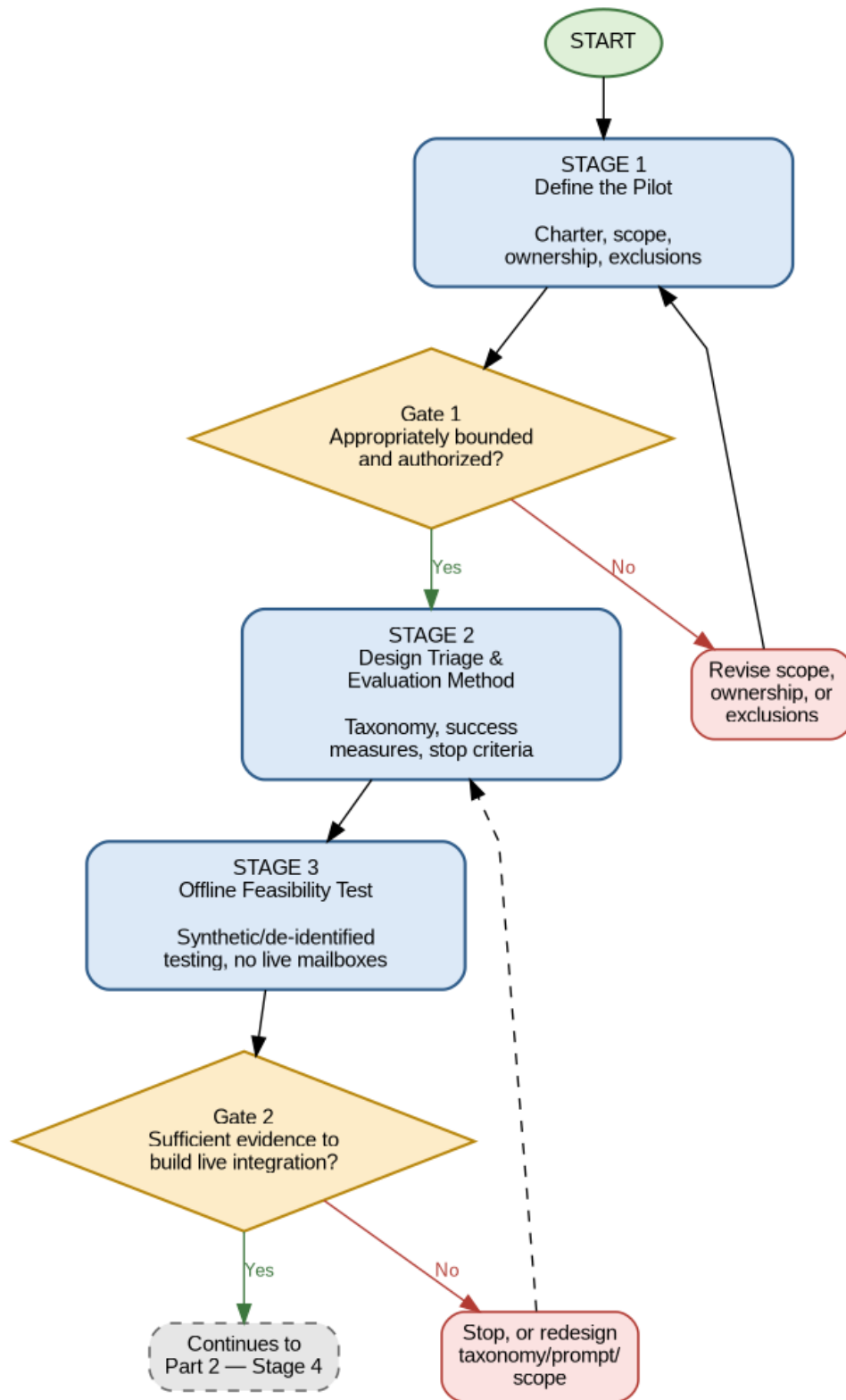


Figure 1 — Stages 1–3: Define the Pilot through Offline Feasibility Test

Workflow Diagram — Part 2 of 5

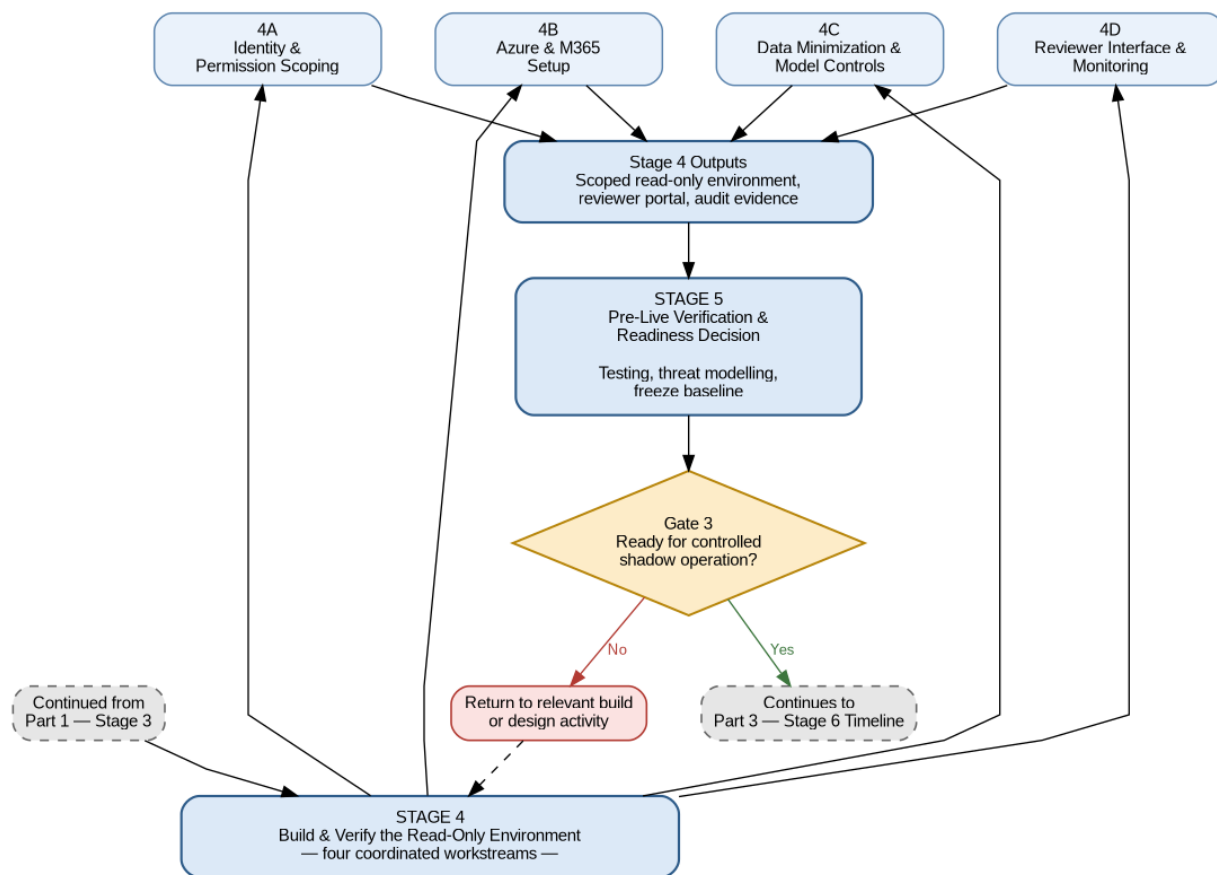


Figure 2 — Stage 4 coordinated workstreams (4A–4D) and Stage 5 readiness decision

Note: 4A–4D are coordinated workstreams that proceed together, not a fixed sequence.

Workflow Diagram — Part 3 of 5

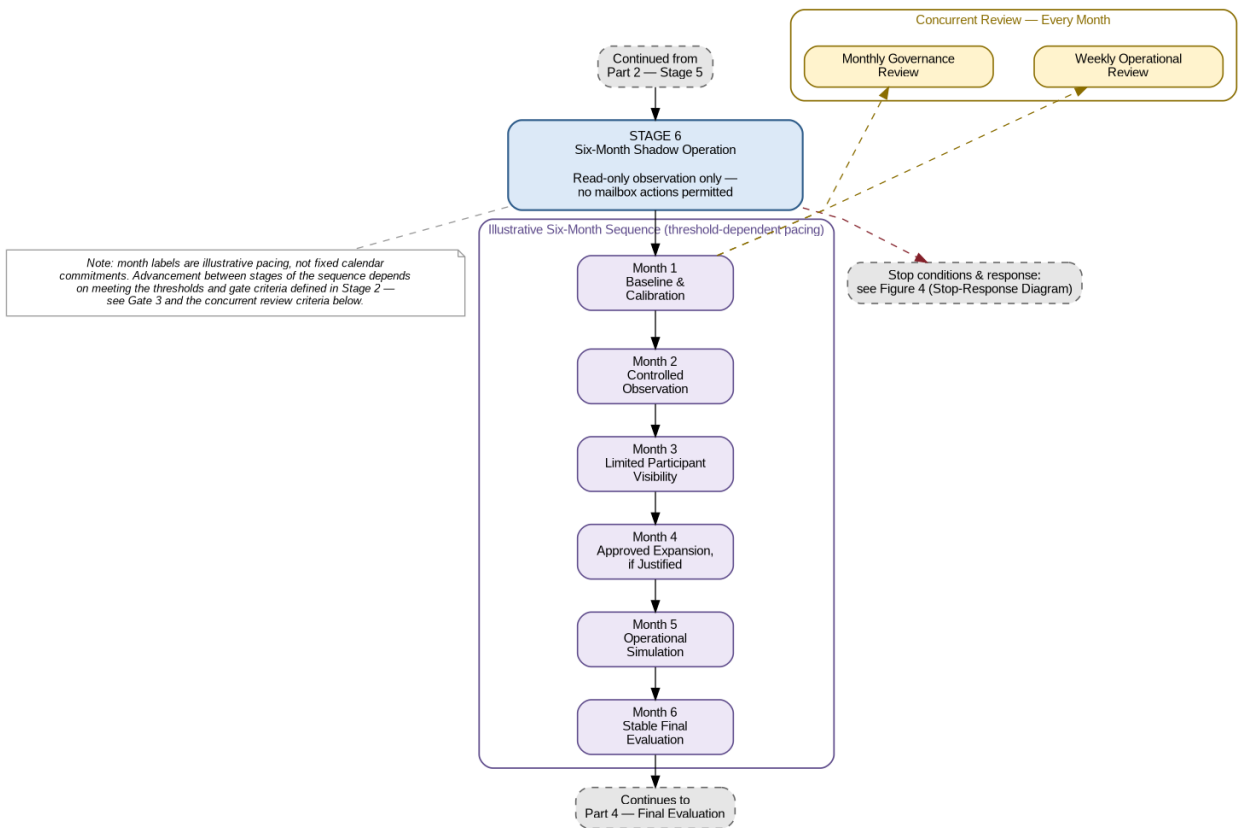


Figure 3 — Stage 6 timeline: illustrative six-month sequence and concurrent review

Note: month labels are illustrative and threshold-dependent — progression is paced by meeting review criteria, not by the calendar.

Workflow Diagram — Part 4 of 5

This diagram applies throughout Stage 6 (all six months) and overrides the timeline in Figure 3 at any point it is triggered.

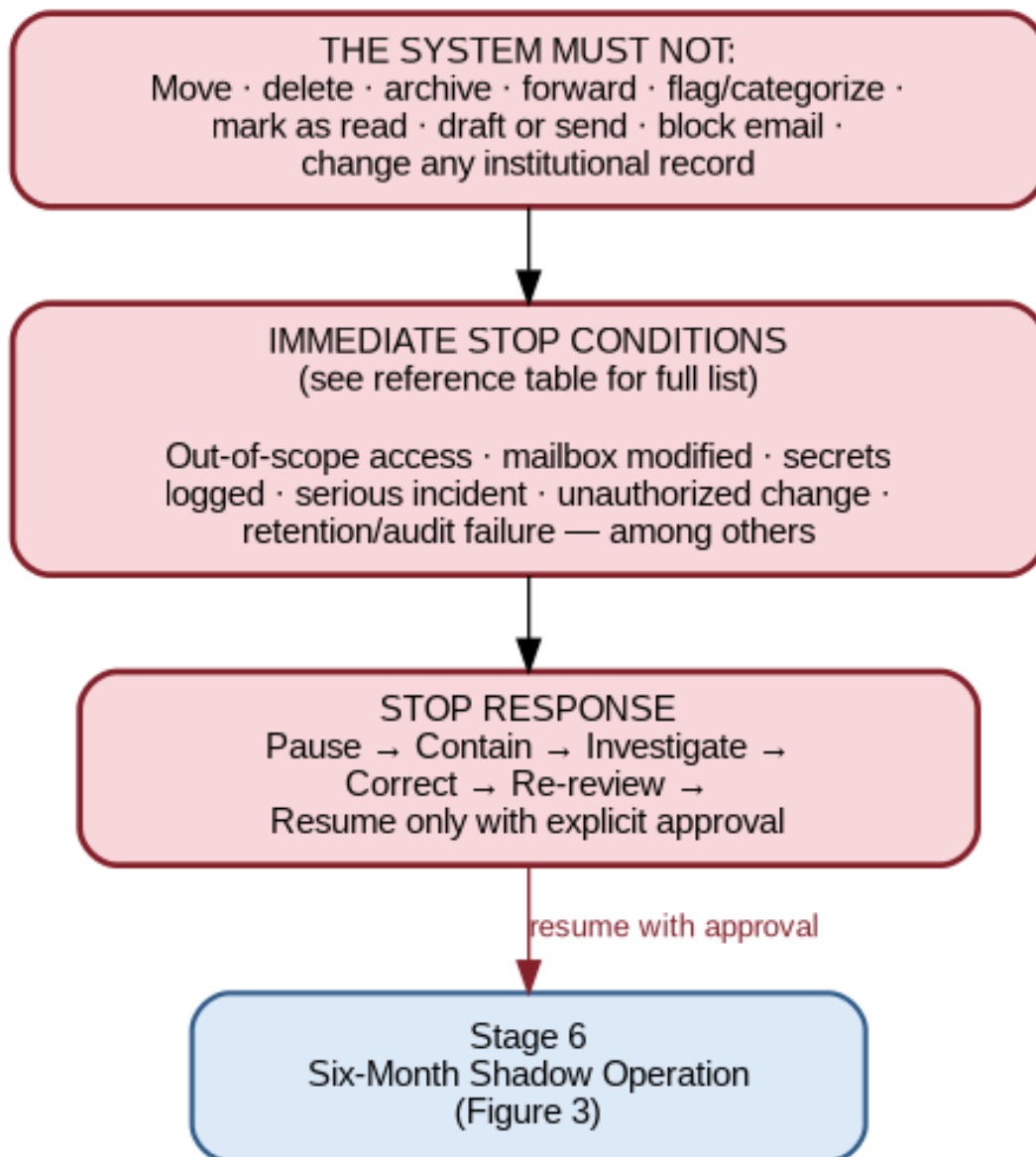


Figure 4 — Stage 6 stop-response diagram (applies throughout the six-month operation)

Workflow Diagram — Part 5 of 5

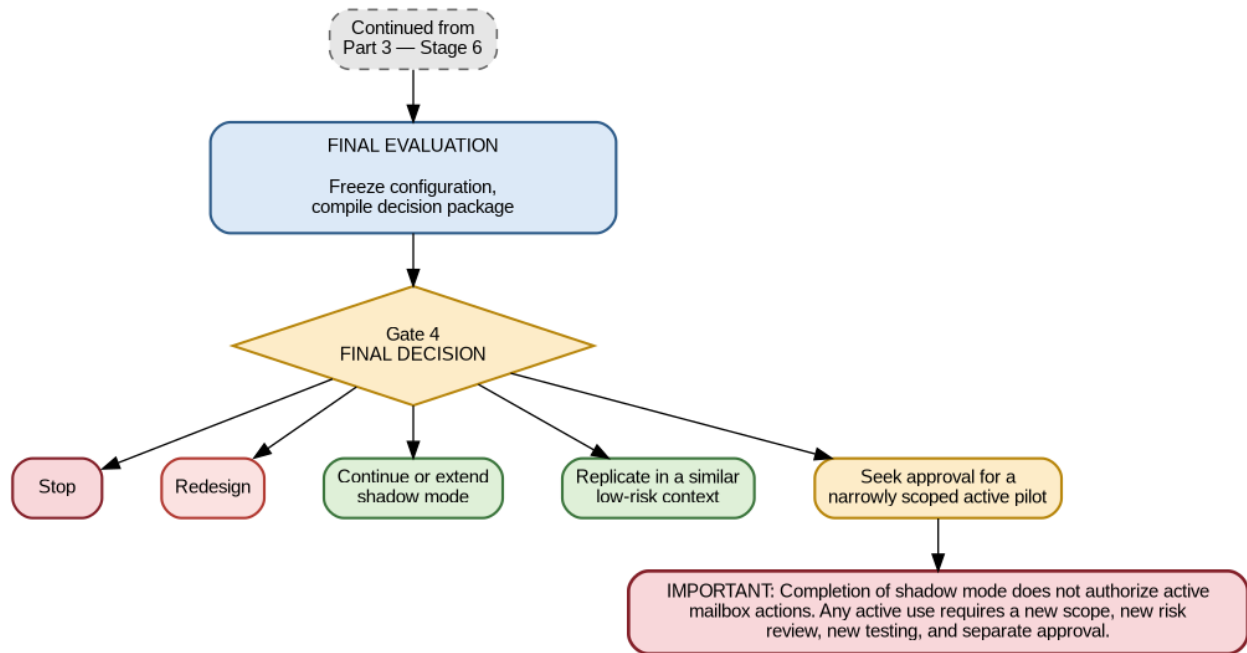


Figure 5 — Final Evaluation and Gate 4: final decision

Detailed Stage Reference

Full activity, ownership, output, and gate logic for each stage, in process order. This is the authoritative detail; the diagrams are navigation aids.

STAGE 1	Define the Pilot	
Activity	Select one approved administrative unit or a small set of closely related mailboxes; define the operational problem; document the current human triage workflow; define the six-month shadow scope; identify participants, reviewers, exclusions, retention, and stop conditions; confirm that the pilot will remain read-only and non-operative.	
Responsible	Operational owner; Pilot coordinator; mailbox subject-matter experts. Specialist input as required: information security, privacy or legal, records management, Microsoft 365 / Exchange administration.	
Outputs	Pilot charter; current-state workflow; approved mailbox scope; defined exclusions; named owner and reviewers; initial approval record.	
Gate 1	Is the proposed pilot appropriately bounded and authorized?	
	If No	Revise scope, ownership, exclusions, or data approach → return to Stage 1.
	If Yes	Proceed to Stage 2.

STAGE 2	Design the Triage & Evaluation Method
Activity	Define a limited triage taxonomy and priority levels, including an explicit “Uncertain” result; document the reviewer workflow; define success measures, safety thresholds, and immediate-stop conditions; establish a human-performance baseline.

STAGE 2	Design the Triage & Evaluation Method
Responsible	Pilot coordinator; mailbox subject-matter experts and reviewers; evaluation lead.
Outputs	Controlled taxonomy; classification definitions; reviewer instructions; baseline measures; evaluation framework; approved safety and stopping criteria.

STAGE 3	Offline Feasibility Test	
Activity	Create a representative synthetic or properly de-identified message set; test the taxonomy and model without connecting to live mailboxes; compare AI recommendations with expert reviewer judgments; test ambiguity, urgent cases, multilingual content, malformed messages, and prompt-injection attempts; assess whether the approach is sufficiently useful to justify technical integration.	
Responsible	Evaluation lead; mailbox subject-matter experts and reviewers; technical implementation team.	
Outputs	Offline test results; taxonomy revisions; initial accuracy and error analysis; recommendation to stop, redesign, or proceed.	
Gate 2	Is there sufficient evidence to justify building the live read-only integration?	
	If No	Stop, or redesign taxonomy, prompt, scope, or workflow → return to Stage 2 or Stage 3.
	If Yes	Proceed to Stage 4.

STAGE 4 — Build and Verify the Read-Only Pilot Environment

The four workstreams below are coordinated, not sequential — they proceed together and converge on the outputs listed after 4D.

4A	Identity, Permission & Mailbox Scoping
Activity	Create pilot, reviewer, and administrator groups; register the Entra application; grant Mail.Read only; do not grant Mail.ReadWrite or Mail.Send; restrict access to approved pilot mailboxes; verify that access to excluded mailboxes fails.

4B	Azure & Microsoft 365 Setup
Activity	Deploy approved Azure resources; configure managed identities, Key Vault, Service Bus, Graph subscriptions, and monitoring/audit services; separate development, test, and pilot environments where required.

4C	Data Minimization & Model Controls
Activity	Retrieve only required fields; prefer body preview over full message body; exclude attachments initially; normalize and redact content; treat all email content as untrusted input; use a constrained prompt and structured output schema; validate all model responses; reject any prohibited mailbox action; store only minimal evaluation data.

4D	Reviewer Interface & Monitoring
Activity	Build the reviewer interface; restrict reviewers to authorized messages; record reviewer decisions and correction reasons; enable audit logging and operational monitoring; exclude message bodies, credentials, and secrets from logs; configure retention and deletion controls.
Responsible (4A–4D)	Technical implementation team; Microsoft 365 / Exchange administrator. Specialist reviews: information security, privacy or legal, records management.
Outputs (4A–4D)	Scoped read-only environment; permission inventory; validated processing pipeline; reviewer portal; audit and monitoring evidence; retention and deletion configuration.

STAGE 5	Pre-Live Verification & Readiness Decision
Activity	Test included and excluded mailbox access; confirm Mail.Read only; test permission changes and access revocation; conduct threat modelling; test prompt-injection and malformed-content handling; test failover, timeout, load, retention, deletion, and emergency stop; process several hundred synthetic or de-identified messages; verify that no mailbox action can occur; complete privacy, security, records, and operational reviews; freeze the approved configuration baseline.
Responsible	Technical implementation team; evaluation lead; specialist reviewers; approving authority.

STAGE 5	Pre-Live Verification & Readiness Decision	
Outputs	Technical test report; privacy and security evidence; permission validation; evaluation baseline; completed readiness checklist.	
Gate 3	Is the system ready for controlled shadow operation?	
	If No	Return to the relevant build or design activity → correct and retest.
	If Yes	Proceed to Stage 6 (six-month shadow operation).

STAGE 6	Six-Month Shadow Operation
Operating boundary	Read-only observation; AI classification and explanation; human comparison and reviewer feedback; no mailbox actions permitted.
The system must not	Move, delete, archive, forward, flag, or categorize email; mark email as read; draft or send email; block email; change any institutional record.
Responsible	Operational owner; pilot coordinator; mailbox subject-matter experts and reviewers; evaluation lead; technical implementation team for operational support.

Six-Month Operating Sequence — Illustrative and Threshold-Dependent

The month labels below describe an illustrative pace, not a committed calendar. Movement from one point in the sequence to the next depends on meeting the success measures, safety thresholds, and stop criteria defined in Stage 2, confirmed through the weekly and monthly review described below. The sequence can pause, repeat, or extend within a step if criteria are not yet met.

Month	Focus & Activity
Month 1	Baseline and calibration — validate taxonomy; establish reviewer agreement; review a high proportion of recommendations; identify data-quality problems.
Month 2	Controlled observation — continue read-only processing; conduct weekly error analysis; limit unnecessary model or prompt changes.
Month 3	Limited participant visibility — show approved recommendations or summaries to selected participants; assess trust, clarity, and usefulness.
Month 4	Approved expansion, if justified — expand only within the authorized scope; examine performance by message type, language, or mailbox.
Month 5	Operational simulation — record “would recommend” or “would route” actions without executing any action; conduct incident and emergency-stop exercises.
Month 6	Stable final evaluation — freeze material configuration changes; collect final evidence; prepare the decision package.

Concurrent Review — Throughout All Six Months

Cadence	Scope
Weekly operational review	Processing failures; subscription status; classification errors; missed and false urgent cases; latency and costs; permission and deployment changes; user complaints; reviewer-access anomalies.

Cadence	Scope
Monthly governance review	Aggregate accuracy; error trends; differential performance; privacy and security findings; records and retention compliance; participant and reviewer lists; continued justification for processing.

Stop Conditions & Response — see Figure 4

Pause processing immediately if:

- A mailbox outside the approved scope is accessed
- Any mailbox is modified
- An email is sent, moved, deleted, forwarded, categorized, flagged, or blocked
- Credentials, access tokens, secrets, or full message bodies appear in logs
- A serious privacy or security incident occurs
- Critical messages are repeatedly missed
- Discriminatory or prohibited inferences occur
- A model, prompt, permission, or deployment changes without approval
- Reviewer access exceeds the approved need
- Retention, legal-hold, deletion, or audit obligations cannot be met

Stop response: Pause → Contain → Investigate → Correct → Re-review → Resume only with explicit approval.

FINAL EVAL.	Final Evaluation	
Activity	Freeze configuration; complete technical report; complete privacy and threat assessment; confirm permission inventory; analyze accuracy and error patterns; analyze differential performance where appropriate; assess staff effort and reviewer burden; assess service impact and cost; summarize complaints and incidents; confirm retention and deletion; compare results with the original baseline.	
Responsible	Evaluation lead; operational owner; supporting technical and specialist reviewers.	
Output	Final decision package.	
Gate 4	Final Decision	
	If No	Stop, or redesign.
	If Yes	Continue or extend shadow mode; replicate in a similar low-risk context; or seek separate approval for a narrowly scoped active pilot.

